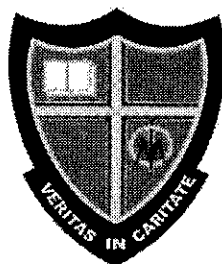
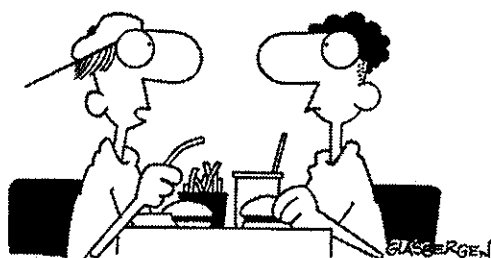


ST BENEDICT'S COLLEGE



SUBJECT	AP Stats	DATE	26 July 2016
GRADE	11	MARKS	100
EXAMINER	Mrs Sillman	MODERATORS	Mrs Povall Mrs Serafino
NAME		DURATION	1.5 hours
CLASS		TEACHER	

QUESTION NO	DESCRIPTION	MAXIMUM MARK	ACTUAL MARK
1 - 3	Conditional Probability	32	
4 - 6	Counting methods	32	
7 - 10	Probability Distributions	36	
		100	



"I forgot to make a back-up copy of my brain,
so everything I learned last semester was lost."

%

INSTRUCTIONS

1. This question paper consists of 5 pages (10 questions). Read the questions carefully.
2. Answer all the questions.
3. You may use an approved non-programmable and non-graphical calculator, unless otherwise stated.
4. Round off your answers to three decimal digits where necessary.
5. All the necessary working details must be clearly shown. Answers only, without relevant calculations, may incur penalties.
6. It is in your own interest to write legibly and to present your work neatly.

QUESTION 1**[13 marks]**

A group of first year students at a local university were surveyed in order to determine the factors that influenced their decision to attend the university. The survey revealed the following:

55 said "it's the only university offering my course" (A)

51 said "it has an excellent party reputation" (B)

61 said "it's the only place that accepted me" (C)

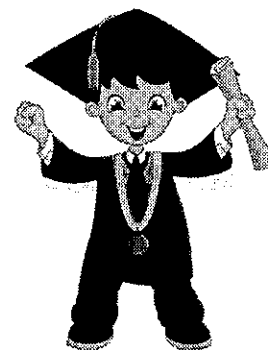
9 said A but did not say B or C

26 said B and A and C

31 said A and B

8 said only B

4 said none of the above



- a) Represent the given information in a Venn diagram. (4)
- b) How many students were surveyed? (2)
- c) Calculate:
 - (i) $P(A \cup B)$ (2)
 - (ii) $P(A \cap B)$ (2)
 - (iii) $P(B|C)$ (3)

QUESTION 2**[8 marks]**

Events H and G are such that:

$$P(G') = 0,3 \quad P(H|G') = 0,5 \quad P(H'|G) = 0,857$$

By drawing a tree diagram or otherwise, find:

- a) $P(H)$ (3)
- b) $P(G \cap H)$ (2)
- c) $P(H \cup G)$ (3)

QUESTION 3**[11 marks]**

An airline wants to determine if passengers not checking luggage is related to people being on business trips. Data for 1000 random passengers at an airport was collected and summarised in the table below:

	<i>Checked baggage</i>	<i>No checked baggage</i>	<i>TOTALS</i>
<i>Travelling for business</i>	103	387	490
<i>Not travelling for business</i>	216	294	510
	319	681	1000

If 'travelling for business' is event A and 'no checked baggage' is event B, answer the following questions:

- a) Find:
- (i) $P(A)$ (1)
 - (ii) $P(B|A)$ (3)
 - (iii) $P(A'|B)$ (3)
- b) Show any calculations to determine if events A and B are independent. (4)

QUESTION 4**[14 marks]**

- a) Expand $n!$ (2)
- b) Simplify ${}^{n+2}P_3$ in terms of n (6)
- c) If ${}^nP_r = 120 \times {}^nC_r$, find r (6)

QUESTION 5**[12 marks]**

A card is drawn at random from an ordinary pack of 52 cards and is not returned to the pack. A second card is drawn at random. Calculate, showing details of your working the probability that:

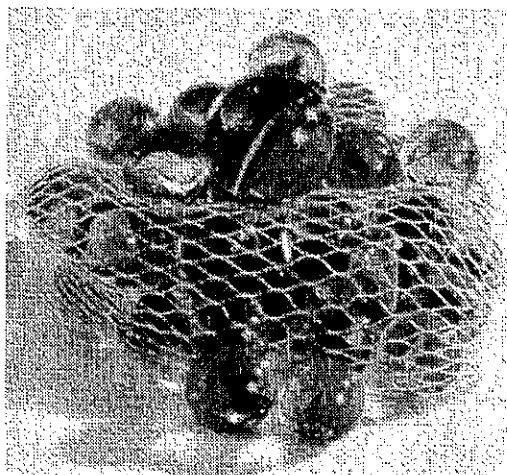
- a) neither card is a club. (3)
- b) the first card is the king of spades and the second card is a spade (3)
- c) the first card is a king and the second card is the ace of the same suit (3)
- d) the first card is a heart and the second is a spade (3)

QUESTION 6**[6 marks]**

- a) How many different 8 letter 'words' can be made from the word BENEDICT? (3)
- b) A four digit number is formed from the digits 1, 2, 3 and 5 and repetition is NOT allowed. Find the probability that the number is divisible by 5.
Explain your answer. (3)

QUESTION 7**[6 marks]**

Calculate the probability that you will get exactly five hearts, including the king of hearts, if you draw 10 cards at random from a regular pack of cards.

QUESTION 8**[10 marks]**

A box contains 12 red and 8 blue marbles. If 5 marbles are picked out at random, what is the probability that 3 reds and two blues will be picked out if:

- a) each marble is replaced before the next one is selected (5)
- b) the marbles are not replaced during the exercise (5)

QUESTION 9**[12 marks]**

A manufacturer of a bag of sweets claims that there is a 90% chance the bag contains some toffees. If 20 bags are chosen, what is the probability that:

- a) all the bags contain toffees
- b) more than 18 bags contain toffees



(5)

(7)

QUESTION 10**[8 marks]**

A twelve person jury is to be selected from 15 men and 14 women.

- a) In how many ways can the jury be formed? (3)
- b) What is the probability that the jury will have exactly 4 men serving on it? (5)